
M.Sc.(AI&ML) - I
Practical Assignment – II
Data Structures using C++

Note :

- (1) Programs should have proper validations for the inputs**
- (2) Use proper variable names**
- (3) Program should contain program definition as comments and should have proper Indentation**

Operator Overloading & Template

- 1) WAP to use binary operator + add two objects of class Numbers having num1 and num2 as its data members and display result.
- 2) WAP to overload operator * which multiply a number to each element of an array within a class arrayContainer and display the result.
- 3) WAP to Overload the *, +, -, ==, != and = operators for the complex class.
- 4) WAP to define an object m1 of matrix class, use m1<<cout.
- 5) WAP to define a matrix class and overload the * operator to multiply a number with matrix (Example: 5*Matrix should be possible).
- 6) WAP to define a class Date with properties int month; int day; int year; overload the following operators.
 - 5.1) + operator [a+b] (a is of date type and b is an integer), use the assumption that all years all years have 360 days and months 30 days.
 - 5.2) - operator [a-b(same as above)]
 - 5.3) = operator
 - 5.4) <, <=, >, >=
 - 5.5) ++, --[post and pre both]
- 7) WAP to define a class Time with properties int hour; int minute; int second; overload the following operators.
 - 6.1) + operator [a+b] (a is of time type and b is an integer)
 - 6.2) - operator [a-b(same as above)]
 - 6.3) = operator
 - 6.4) <, <=, >, >=
 - 6.5) ++, --[post and pre both]

- 8) Write a menu driven program that can perform the following functions on strings. (Use overloaded operators where possible).(Do not use predefined string function or class.)
1. Compare two strings for equality (== operator)
 2. Check whether first string is smaller than the second (<= operator)
 3. Copy the string to another
 4. Extract a character from the string (Overload [])
 5. Reverse the string
 6. Concatenate two strings (+ operator)
- 9) Write an object-oriented program to implement a generic Number Class that can accept either int or float data type and perform basic calculation like +,-,/ and *.
- 10) Write a generic function that will sort a character string, integer and float value. Create a menu with appropriate options and accept the values from the user.
- 11) Write a template function called find(). This function searches an array for an object. It returns either the index of the matching object (if one is found) or -1 if no match is found.